

Supplementary Material for: A Reproducible Software Workflow for Compact Pauli Analysis of Molecular Vibrational Hamiltonians

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I. ADDITIONAL REPRESENTATIVE SPECTRA

II. STAGE-RESOLVED PATULIN BENCHMARK

III. SEPARATE THREE-MOLECULE END-TO-END TIMING CONTROL

A prior integrated validation script (different from the final Pauli-compression code) produced the following timing controls. They are retained only to prevent unsupported speedup claims and should not be merged numerically with the final-code benchmark without rerunning all molecules under the same implementation and hardware conditions.

IV. PROTEIN SCALING STRESS TEST

A separate 1CRN protein benchmark contained 642 atoms and 1926 Cartesian degrees of freedom. A matrix-free reduced Krylov calculation used a primary 64-

dimensional subspace (6 reduced qubits), compared with an 11-qubit full-space index-register count. The run reported a 90.03% reduction in the number of force calls relative to the finite-difference full-Hessian estimate, 28.30 MiB of full Hessian storage versus 0.0312 MiB for the reduced operator, and a same-reduced-operator ST4 $r = 2$ spectral cosine of 0.99999998. However, Dense-NMA versus the primary Lanczos spectrum had cosine 0.8643. This result is therefore a scalability stress test, not evidence that the current 64-dimensional reduction replaces full protein NMA. Rank convergence against the dense reference is required before such a claim.

V. FILES INCLUDED WITH THIS PROJECT

The archive includes the current benchmark code snapshot, the ten-molecule manifest, derived resource table, Patulin Pauli-compression CSV, Patulin Trotter-convergence CSV, and Patulin stage-resolved timing CSV. These machine-readable files should be deposited in a public repository and assigned a DOI before submission.

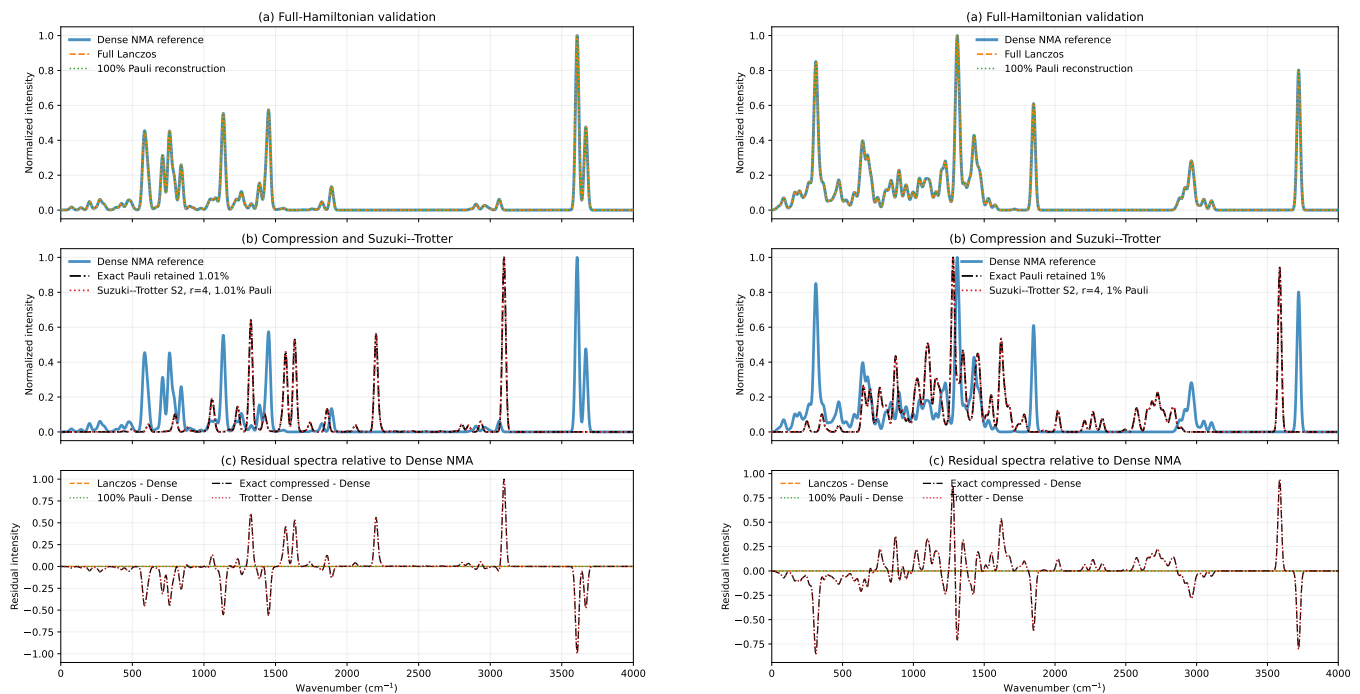


FIG. 1. Citrinin (left) and T-2 toxin (right) full-Hamiltonian, 1% compression/Trotter, and residual panels.

TABLE I. Measured Patulin wall time and RSS changes from the current final-code smoke test. RSS delta is a process-level measurement, not a peak-allocation profiler.

Stage	Time (s)	RSS before (MiB)	RSS after (MiB)
3D preparation/optimization	0.02084	199.55	207.95
Finite-difference Hessian	0.01862	208.41	208.41
Dense NMA	0.000576	208.41	209.43
Full Lanczos	0.001188	210.00	210.46
Low-mode eigsh	0.001823	210.46	210.96
Normalize/pad	0.000163	211.43	211.43
Pauli decomposition	0.03994	211.43	211.73
100% Pauli reconstruction	0.03336	211.73	211.98
10% Pauli reconstruction	0.003476	213.36	213.45
1% Pauli reconstruction	0.000504	213.50	213.58
Exact exp., 1% same H	0.000188	236.80	236.95
ST2, 1%, $r = 1$	0.001352	237.61	237.61
ST2, 1%, $r = 2$	0.002550	238.03	238.03

TABLE II. Historical timing control from the uploaded run log. The reported speedup metric is below unity in all cases; therefore the current architecture does not demonstrate end-to-end speedup over classical NMA.

Molecule	NMA eig. (s)	Krylov (s)	ST4 $r = 2$ (s)	Speedup metric
Anatoxin-a	0.01119	1.7069	0.7408	0.2515
Aflatoxin B1	0.00546	1.5135	0.9197	0.3523
Curcumin	0.00813	1.9554	2.4023	0.3245

VI. Quantum-resource viewer diagnostics

The extended viewer is designed to expose only quantities that are available from validated numerical outputs. Table 1 distinguishes implemented or directly derivable quantities from quantities that require additional quantum-state calculations.

Table 1: Interpretation rules for the extended viewer.

Viewer element	Source quantity	Allowed interpretation
Compact qubit count	$n_q = \lceil \log_2 D \rceil$	Index-register size for the displayed operator or block
Binary basis label	Integer basis index b	Compact basis-state label; not an atom-to-qubit assignment
Pauli-term graph	Retained $c_j P_j$ terms	Operator coupling / shared Pauli support
Trotter fidelity	Same-H exact and product-formula outputs	Product-formula accuracy for the same retained Hamiltonian
Runtime panel	Stage-resolved timing CSV	Classical measured wall time for named pipeline stages
Protein motion	Stored Cartesian eigenvectors	Structural displacement pattern for the selected normal mode
Entanglement	Not present in current outputs	Must remain disabled or explicitly labeled illustrative until state-derived metrics are computed
Quantum speedup	Not established	Must not be displayed as a validated result

Basis-index mapping

For an operator or modal block of dimension $D = 2^{n_q}$, the viewer displays each basis-state index as an n_q -bit string. For $D = 64$, indices 0 through 63 correspond to $|000000\rangle$ through $|111111\rangle$. If the physical block has fewer than D states, padded labels are identified separately and are not presented as physical modes.

Qubit-coupling matrix

Let $s_j(a) = 1$ when Pauli string P_j acts nontrivially on qubit a and zero otherwise. A simple coupling score used only for visualization is

$$C_{ab} = \sum_{j=1}^K |c_j| s_j(a) s_j(b), \quad a \neq b. \quad (1)$$

A normalized variant $\tilde{C}_{ab} = C_{ab} / \max_{a < b} C_{ab}$ can control graph edge width. This quantity measures shared support in the retained Hamiltonian. It is not concurrence, mutual information, entropy, or any other entanglement measure.

What would be required for an entanglement panel

A genuine entanglement panel requires a propagated state $|\psi(t)\rangle$ or density operator $\rho(t)$. For a subsystem A , one may compute

$$\rho_A(t) = \text{Tr}_{\bar{A}} \rho(t), \quad S_A(t) = -\text{Tr} [\rho_A(t) \log_2 \rho_A(t)]. \quad (2)$$

Pairwise mutual information could similarly be displayed as

$$I(a : b) = S(\rho_a) + S(\rho_b) - S(\rho_{ab}). \quad (3)$$

These quantities are not inferred from Hamiltonian connectivity. They should be added only after the software stores the propagated state and validates the reduced-density-matrix calculation.

VII. Protein viewer mapping example and limitations

The 1CRN stress test in the present Supplementary Material contains 642 atoms and 1926 Cartesian degrees of freedom. Its primary reduced Krylov space has dimension 64, corresponding to six reduced index qubits, while the

full-space index-register count is 11. The reported reduced-space ST4 spectrum closely tracks the same reduced operator, but the Dense-NMA versus primary Lanczos spectral cosine is 0.8643. Therefore the example is useful for testing visualization, memory accounting, and mode-to-register mapping, but it does not validate six qubits as a replacement for the complete protein vibrational problem.

A viewer test for this dataset should display at least: (i) structure atom count; (ii) eigenvector atom count and a MATCH/MISMATCH flag; (iii) selected mode and frequency; (iv) reduced-space dimension; (v) compact qubit count; (vi) binary basis label; (vii) retained Pauli-term statistics if available; (viii) same-H Trotter fidelity; and (ix) stage-resolved classical timings. Structural arrows should be calculated from the stored Cartesian displacement vector and normalized only for display.

VIII. Recommended software export fields

To make the viewer reproducible, the numerical backend should export a small JSON record for each dataset with fields such as `physical_dimension`, `padded_dimension`, `n_qubits`, `mode_index`, `frequency_cm1`, `pauli_term_count`, `retained_term_count`, `retained_fraction`, `trace_fidelity`, `unitary_error`, and stage-resolved timings. Protein datasets should additionally record `structure_atom_count`, `eigenvector_atom_count`, and the mode/block association used by the viewer.

The visualization code should treat these files as read-only scientific inputs. A screenshot or animation is then reproducible from the same machine-readable record that supports the manuscript tables.